# Interpretable Machine Learning

## Permutation Feature Importance (PFI)

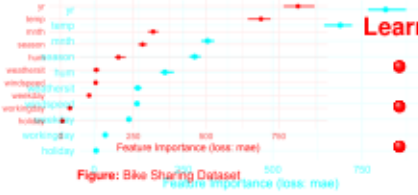


**Figure:** Bike Sharing Dataset

### Learning goals

- Understand how PFI is computed
- Understanding strengths and weaknesses
- Testing Importance

# PERMUTATION FEATURE IMPORTANCE (PFI)

Breiman (2001)

**Idea:** "Destroy" feat. of interest $x_j$ by perturbing it s.t. it becomes uninformative, e.g., randomly permute obs in $x$ (marginal distribution $\mathbb{P}(x_j)$ stays the same).

PFI for features $x_S$ using test data $\mathcal{D}$:

- Measure the error **without permuting feat.** and **with permuted feat. values** $\tilde{x}_S$
- Repeat permuting the feat. (e.g., $m$ times) and avg. the difference of both errors:

$$\widehat{PFI}_S = \frac{1}{m}\sum_{k=1}^{m} \mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}^S_{(k)}) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}),$$

where $\mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}) = \frac{1}{n}\sum_{(x,y)\in\mathcal{D}} L(\hat{f}(x), y)$



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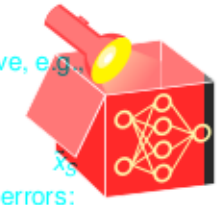

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The data $\mathcal{D}$ where $x_S$ is replaced with $\tilde{x}^S$ is denoted as $\tilde{\mathcal{D}}^S$.

Example of permuting feature $x_S$ with $S = \{1\}$ and $m = 6$:

Note: The $S$ in $x_S$ refers to a **S**ubset of features for which we are interested in their effect on the prediction.

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# PERMUTATION FEATURE IMPORTANCE

$\tilde{\mathcal{D}}^S_{(k)}$ $\mathcal{D}$ $\mathcal{D}$

| $i$ | $x_S$ | $x_1$ | $x_2$ | $x_3$ | $x_1$ | $x_2$ | $x_3$ | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 4 | 7 | 4 | 1 | 4 | 7 | 1 | 4 | 7 |
| ⋮ | 1 | 5 | 8 | 5 | 2 | 5 | 8 | 2 | 5 | 8 |
| n | 3 | 6 | 9 | 6 | 3 | 6 | 9 | 3 | 6 | 9 |

1. **Perturbation:** Sample feature values from the distribution of $x_S$ ($P(X_S)$).
   ⇒ Randomly permute feature $x_S$
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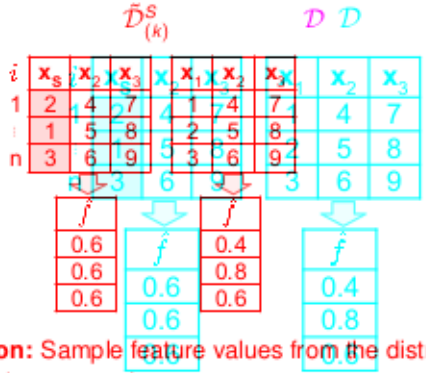

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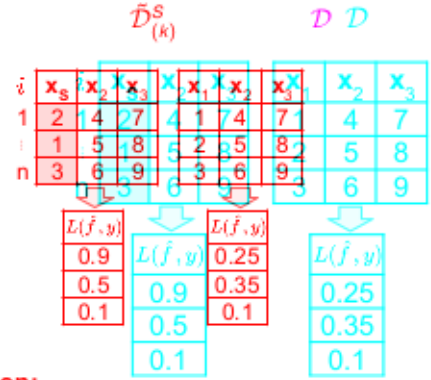

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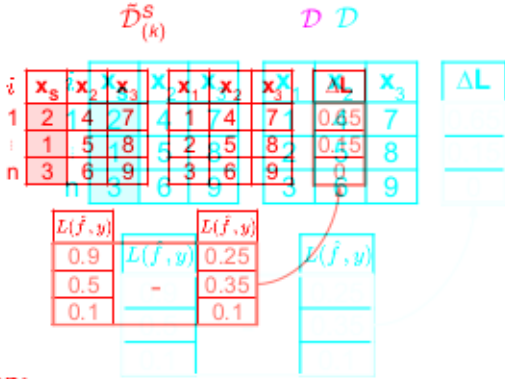

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# PERMUTATION FEATURE IMPORTANCE

$$\mathcal{R}_{emp}(\hat{f}, \tilde{\mathcal{D}}^{s}_{(k)}) - \mathcal{R}_{emp}(\hat{f}, \mathcal{D})$$

| $i$ | $x_s$ | $x_2$ | $x_3$ | $x_1$ | $x_2$ | $x_3$ | $\Delta L$ | $x_3$ | $\Delta L$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 4 | 7 | 1 | 4 | 7 | 0.65 | 7 | 0.65 |
| i | 1 | 5 | 8 | 2 | 5 | 8 | 0.15 | 8 | 0.675 |
| n | 3 | 6 | 9 | 3 | 6 | 9 | 0 | 9 | 0 |

= 0.267

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   - Compute the loss for each observation in both data sets
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   - Average this change in loss across all observations
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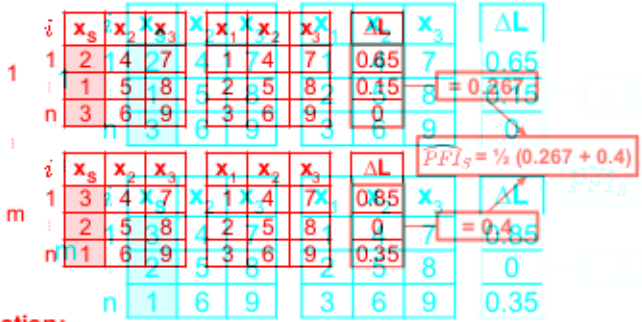


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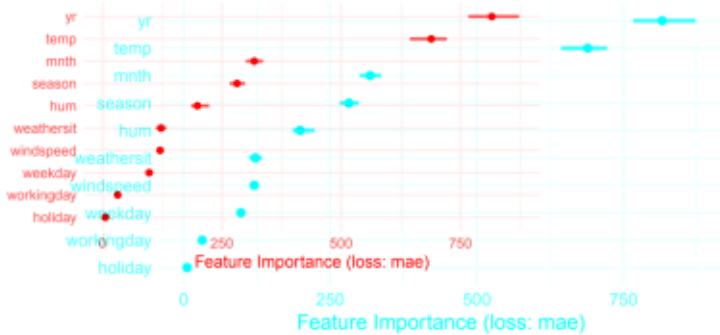
# EXAMPLE: BIKE SHARING DATASET



**Interpretation:**

- Year (yr) and Temperature (temp) are most important features
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- Interpretation: PFI is the increase of model error when feature's information is destroyed



Interpretable Machine Learning – 4 / 12

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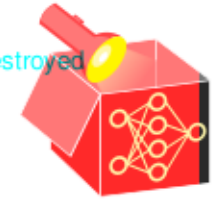



Interpretable Machine Learning – 4 / 12

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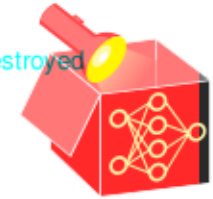

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  - PFI automatically includes importance of interaction effects with other features
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    - ⇒ Importance of all interactions with the permuted feature are contained in PFI score



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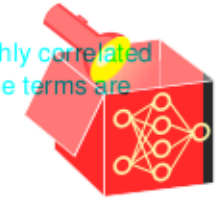
# COMMENTS ON PFI - EXTRAPOLATION

**Example:** Let $y = x_3 + \epsilon_y$ with $\epsilon_y \sim N(0, 0.1)$ where $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$ are $\epsilon_2$ are highly correlated highly correlated ($\epsilon_1 \sim N(0, 1)$, $\epsilon_2 \sim N(0, 0.01)$) and $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim N(0, 1)$. All noise terms are independent. Fitting a LM yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$.





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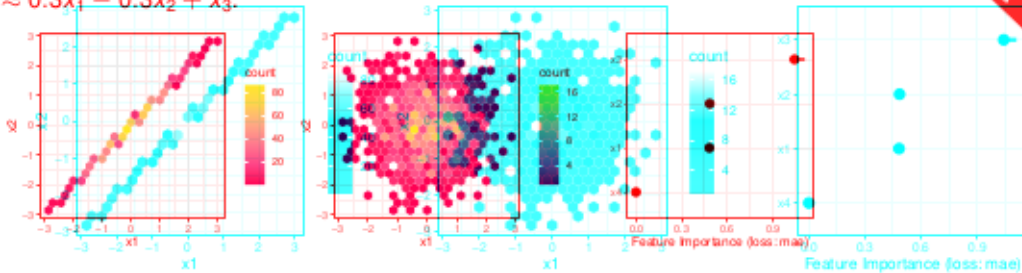


Hexbin plot of $x_1, x_2$ before permuting $x_1$ (left), after permuting $x_1$ (center), and PFI scores (right)

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Interpretable Machine Learning – 5 / 12

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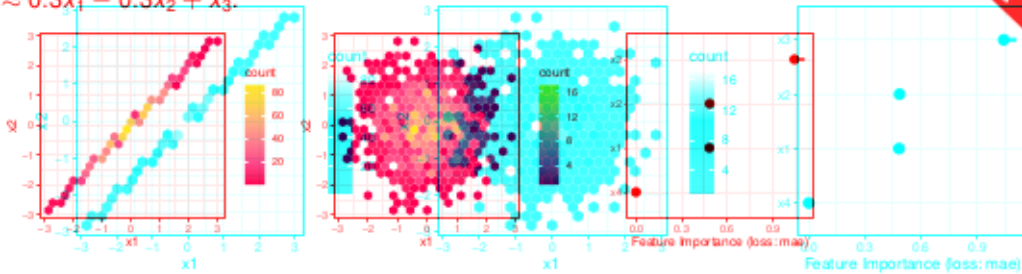


Hexbin plot of $x_1, x_2$ before permuting $x_1$ (left), after permuting $x_1$ (center), and PFI scores (right) $\Rightarrow$ $x_1$ and $x_2$ should be irrelevant for the prediction $\hat{f}(\mathbf{x})$ for $\{\mathbf{x} : \mathbb{P}(\mathbf{x}) > 0\}$ as $0.3x_1 - 0.3x_2 \approx 0$

$\Rightarrow$ PFI evaluates model on unrealistic obs. outside $\mathbb{P}(\mathbf{x})$ $\rightsquigarrow$ $x_1, x_2$ are considered relevant (PFI $> 0$)

# COMMENTS ON PFI - INTERACTIONS

**Example:** Let $x_1, x_2, \ldots, x_4$ be independently and uniformly sampled from $\{-1, 1\}$ and

$$y := x_1 x_2 + x_3 + \epsilon_Y \text{ with } \epsilon_Y \sim N(0, 1)$$

Fitting a LM yields $\hat{f}(x) \approx x_1 x_2 + x_3$.

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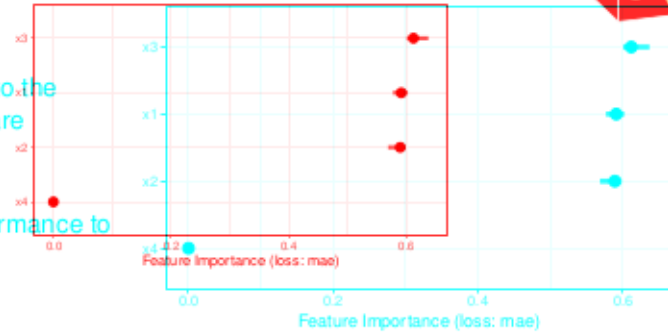
Although $x_3$ alone contributes as much to the prediction as $x_1$ and $x_2$ jointly, all three are considered equally relevant.

$\Rightarrow$ PFI does not fairly attribute the performance to the individual features.



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# COMMENTS ON PFI - TEST VS. TRAINING DATA

**Example:** $x_1, \ldots, x_{20}, y$ are independently sampled from $\mathcal{U}(-10, 10)$. An `xgboost` model with default hyperparameters is fit on a small training set of 50 observations. The model overfits heavily.

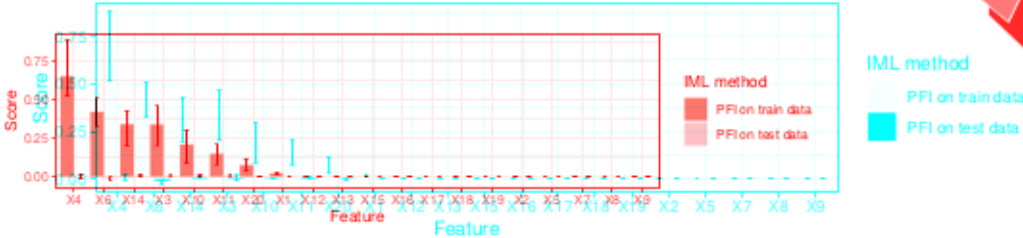


**Figure:** While PFI on test data considers all features to be irrelevant, PFI on train data exposes the features on which the model overfitted.

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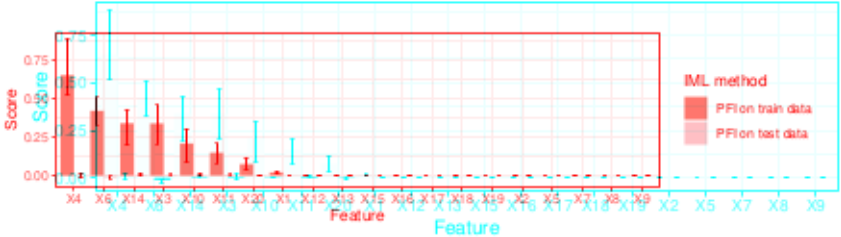


**Figure:** While PFI on test data considers all features to be irrelevant, PFI on train data exposes the features on which the model overfitted.

Why? PFI can only be nonzero if the permutation breaks a dependence in the data. Spurious correlations help the model perform well on train data but are not present in the test data.

⇒ If you are interested in which features help the model to generalize, apply PFI on test data.

# IMPLICATIONS OF PFI

Can we get insight into whether the ...

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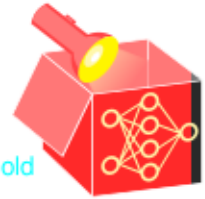
   - As the training vs. test data example demonstrates, the converse does not hold





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2. feature $x_j$ contains prediction-relevant information?
   - $PFI_j = 0 \Rightarrow x_j$ is dependent of $y$ or its covariates $x_{-j}$ or both (due to extrapolation)
   - $PFI_j \neq 0 \Rightarrow x_j$ is dependent of $y$ or it's covariates $x_{-j}$ or both (due to extrapolation)
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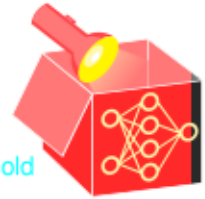

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3. model requires access to $x_j$ to achieve it's prediction performance?
   - $x_j$ is not exploited by model (regardless of whether it is useful for $y$ or not) $\Rightarrow PFI_j = 0$
   - As the extrapolation example demonstrates, such insight is not possible



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# TESTING IMPORTANCE (PIMP)

Altmann et al. (2010)

- PIMP was originally introduced for random forest's built-in permutation feature importance



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  - ⇒ By computing PFI scores again, we obtain distribution of PFI scores under $H_0$
- Compute p-value - the tail probability under $H_0$ - and use it as a new importance measure



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# TESTING IMPORTANCE (PIMP)

PIMP algorithm:

1. For $m \in \{1, 1, \ldots, n_{repetitions}\}$:
   - Permute response vector $y$
   - Retrain model with data $\mathbf{X}$ and permuted $y$
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# TESTING IMPORTANCE (PIMP)

PIMP algorithm:

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2. Train model with $\mathbf{X}$ and unpermuted $y$
3. For each feature $j \in \{1, \ldots, p\}$:
   - Fit probability distribution of the feature importance values $PFI_j^m$, $m \in \{1, \ldots, n_{repetitions}\}$ (choice between Gaussian, lognormal, gamma or non-parametric)
   - Compute feature importance $PFI_j$ for the model without permutation of $y$ (under $H_1$)
   - Retrieve the p-value of $PFI_j$ based on the fitted distribution



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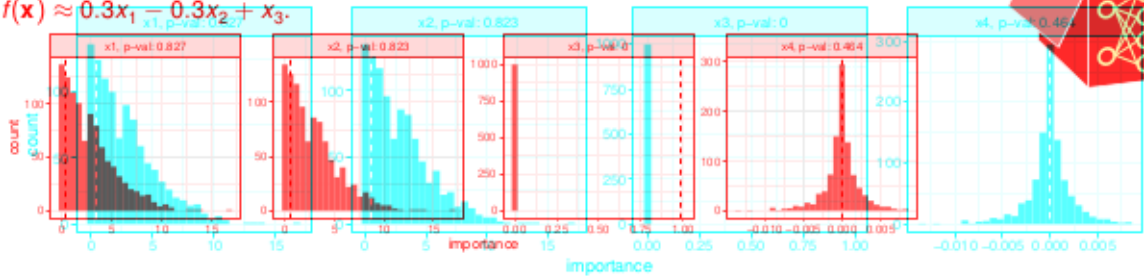
# PIMP FOR EXTRAPOLATION EXAMPLE

**Recall:** $y = x_3 + \epsilon_y$ with $\epsilon_y \sim N(0, 0.1)$, $x_1, x_2$ highly correlated but independent of $y$, $x_4$ is independent of $y$ and all other variables. Fitting a LM yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$.




- Histograms: $H_0$ distribution of PFI scores after permuting $y$ (1000 repetitions)
- Red: PFI score estimated on unpermuted $y$ (under $H_1$) $\rightsquigarrow$ compare against $H_0$ distribution
- Results: Although PFI for $x_1$ and $x_2$ is nonzero (red), PIMP considers them not significantly relevant (p-value > 0.05)

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